📌 CodeIgniter 4 Syntax Cheat Sheet

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1. Load Helpers

Php Copy code

```php
helper('url');        // Load single helper
helper(['form','url']); // Load multiple
```

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2. Routing

app/Config/Routes.php

Php Copy code

```php
$routes->get('/', 'Home::index');        // GET request
$routes->post('submit', 'Form::submit');   // POST
request
$routes->match(['get','post'], 'login', 'Auth::login'); //
Both GET & POST
$routes->get('users/(:num)', 'Users::view/$1'); //
Route with ID param
```

3. Controllers

Create Controller: app/Controllers/Users.php

Php Copy code

```php
namespace App\Controllers;

class Users extends BaseController
{
    public function index()
    {
        return view('users_list');
    }

    public function show($id)
    {
        echo "User ID: " . $id;
    }
}
```

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4. Views

Load View:

Php Copy code

```php
return view('users_list'); // app/Views/users_list.php
```

Pass Data to View:

Php Copy code

```php
$data = ['title' => 'My Users'];
return view('users_list', $data);
```

View File:

Php Copy code

```php
<h1><?= $title ?></h1>
```

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5. Models

Create Model: app/Models/UserModel.php

Php Copy code

```php
namespace App\Models;
use CodeIgniter\Model;

class UserModel extends Model
{
    protected $table = 'users';
    protected $primaryKey = 'id';
    protected $allowedFields = ['name', 'email',
'password'];
}
```

6. Database Operations

Php Copy code

```php
$userModel = new \App\Models\UserModel();

// Insert
$userModel->save([
    'name' => 'John Doe',
    'email' => 'john@example.com'
]);

// Get All
$users = $userModel->findAll();

// Get One
$user = $userModel->find($id);

// Update
$userModel->update($id, ['email' =>
'new@example.com']);

// Delete
$userModel->delete($id);
```

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7. Form Handling

Form in View:

Php Copy code

```php
<form method="post" action="/register">
    <?= csrf_field() ?>
    <input type="text" name="name">
    <button type="submit">Submit</button>
</form>
```

Form Validation in Controller:

Php Copy code

```php
if ($this->request->getMethod() === 'post') {
    $rules = [
        'name' => 'required|min_length[3]',
        'email' => 'required|valid_email'
    ];
    if (! $this->validate($rules)) {
        return view('register', ['validation' =>
$this->validator]);
    }
}
```

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8. Session

Php Copy code

```php
session()->set('user_id', 1);        // Set
$id = session()->get('user_id');     // Get
session()->remove('user_id');        // Remove
session()->destroy();               // Destroy all
```

9. Flash Data

Php Copy code

```php
session()->setFlashdata('success', 'User added successfully.');
echo session()->getFlashdata('success');
```

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10. File Upload

Php Copy code

```php
$file = $this->request->getFile('userfile');
if ($file->isValid() && ! $file->hasMoved()) {
    $file->move(WRITEPATH . 'uploads');
}
```

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11. Password Hashing

Php Copy code

```php
$hash = password_hash('mypassword', PASSWORD_DEFAULT);
if (password_verify('mypassword', $hash)) {
    echo "Password matches";
}
```

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12. Security

Php Copy code

```php
// CSRF field in forms
<?= csrf_field() ?>

// XSS protection
$clean = esc($data);
```

13. Redirect

Php Copy code

```php
return redirect()->to('/dashboard');
```

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14. URL Helpers

Php Copy code

```php
base_url('path/to/file'); // Full URL
site_url('controller/method');
```

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15. Pagination

Php Copy code

```php
$pager = \Config\Services::pager();
$data['users'] = $userModel->paginate(10);
$data['pager'] = $userModel->pager;

<?= $pager->links() ?>
```